

Smoother Action Chunking Flow Policy via Prior-Corrected Orthogonal Trust-Region Guidance

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Abstract—Flow-matching robot policies commonly use action-chunking inference for efficient closed-loop control, but chunk boundaries can introduce discontinuous action transitions. Existing RTC guidance improves continuity by injecting correction signals during denoising, yet its weight schedule is weak at intermediate timesteps and its unconstrained correction direction may introduce transverse perturbations. We propose POTR, a prior-corrected orthogonal trust-region guidance method. First, we incorporate a data-prior scale σ_d into the RTC guidance weight, yielding stronger intermediate-time correction. Second, we decompose the guidance vector into components parallel and perpendicular to the denoising velocity, and constrain the perpendicular component within a trust region. On LIBERO with $\pi_{0.5}$, POTR improves success rate and consistently reduces chunk-boundary discontinuity, acceleration, and jerk compared with RTC. Ablations show that the prior-corrected weight provides the main correction gain, while the orthogonal trust region further improves stability.

Index Terms—flow-matching policy, real-time correction guidance, prior-corrected weight, orthogonal trust region, robot manipulation, trajectory smoothness

I. INTRODUCTION

Flow matching [1], [2] enables robot manipulation policy learning with straighter sampling trajectories than DDPM [3]. π_0 [4] applies flow matching to VLA models; Diffusion Policy [5] combines DDPM with chunked prediction. Asynchronous action-chunking inference [6] improves efficiency but introduces L2 jumps at chunk boundaries, causing acceleration and jerk spikes that degrade safety [7].

The Real-Time Chunking (RTC) method [8] injects observation-guided correction during denoising, building on Π GDM [9] and inpainting [10]. The EDM framework [11] provides a theoretical basis for data scale in guidance. However, RTC has two limitations: (1) the weight at $\tau \approx 0.5$ is only ~ 2.0 , leaving intermediate denoising under-corrected; (2) the correction direction is unconstrained, producing perpendicular perturbations at chunk boundaries. Alternatives like BID [12] and Streaming Diffusion Policy [13] require architectural changes, whereas our method modifies only the guidance signal.

We propose POTR with two components: (1) a **prior-corrected weight** using σ_d to boost intermediate guidance by $\sim 3.6\times$; (2) an **orthogonal trust-region constraint** clipping the perpendicular component ($\|\mathbf{g}'_{\perp}\| \leq \rho\|\mathbf{v}_{\tau}\|$). We evaluate

on LIBERO [14] with 6 metrics spanning success rate, efficiency, and smoothness.

II. PRELIMINARIES AND MOTIVATION

A. Flow Matching and Action Chunking

Consider a flow-matching policy π_{θ} with asynchronous chunked inference: every s steps the model generates an action chunk $\mathbf{A}_t = [a_t, \dots, a_{t+H-1}]$ of length H . The conditional flow matching path is:

$$p_{\tau}(\mathbf{A}_t^{\tau} | \mathbf{A}_t^1) = \mathcal{N}(\tau \mathbf{A}_t^1, (1-\tau)^2 \mathbf{I}) \quad (1)$$

where $\tau \in [0, 1]$ is the denoising timestep. Discretization via n -step Euler solver:

$$\mathbf{A}_t^{\tau+1/n} = \mathbf{A}_t^{\tau} + \frac{1}{n} v_{\pi}(\mathbf{A}_t^{\tau}, \mathbf{o}_t, \tau) \quad (2)$$

B. RTC Guidance Framework

RTC [8] injects residual actions from the previous chunk as an inpainting [10] target at each denoising step, building on Π GDM [9]:

$$\mathbf{v}_{\Pi\text{GDM}} = \mathbf{v}_{\pi} + \min\left(\beta, \frac{1-\tau}{\tau \cdot r_{\tau}^2}\right) \cdot (\mathbf{Y} - \hat{\mathbf{A}}_t^1)^{\top} \text{diag}(\mathbf{W}) \frac{\partial \hat{\mathbf{A}}_t^1}{\partial \mathbf{A}_t^{\tau}} \quad (3)$$

where $\hat{\mathbf{A}}_t^1 = \mathbf{A}_t^{\tau} + (1-\tau) \mathbf{v}_{\pi}$ is the one-step clean estimate, $\mathbf{Y}$ is the inpainting target (residual actions from previous chunk), $\mathbf{W} \in \mathbb{R}^H$ is the soft mask, $r_{\tau}^2 = (1-\tau)^2/(\tau^2 + (1-\tau)^2)$, and β is the clipping threshold. The guidance weight is $w_{\text{RTC}}(\tau) = \min((1-\tau)/(\tau \cdot r_{\tau}^2), \beta)$. Defining $\text{SNR}(\tau) = \tau^2/(1-\tau)^2$, this expands to $w_{\text{RTC}}(\tau) = \min((1-\tau)(1+\text{SNR}(\tau))/\tau, \beta)$.

The weight forms a bimodal structure: at $\tau \rightarrow 0$ and $\tau \rightarrow 1$ it is high, but at the midpoint $\tau \approx 0.5$ it reaches its minimum ($r^{-2} = 2$) — precisely where the intermediate denoising phase often plays an important role in shaping the generated action structure. This mismatch is the primary limitation of RTC.

III. METHOD

A. Prior-Corrected Weight Function

RTC and Π GDM [9] default to $\sigma_d = 1$ (unit prior variance), assuming pixel-normalized image generation. In robot action generation, the observation-conditioned prior $p_1(\mathbf{A}_t^1 | \mathbf{o}_t)$ has

much smaller variance ($\sigma_{d|o}^2 \ll 1$): given a current observation, plausible trajectories are highly concentrated. Retaining σ_d in the weight derivation gives the corrected normalization:

$$r_\tau^2 = \frac{(1-\tau)^2 \sigma_d^2}{(1-\tau)^2 + \sigma_d^2 \tau^2} \quad (4)$$

which reduces to RTC's $r_\tau^2 = (1-\tau)^2/(\tau^2 + (1-\tau)^2)$ at $\sigma_d = 1$. The prior-corrected weight is:

$$w_{PC}(\tau) = \min\left(\frac{(1-\tau)^2 + \sigma_d^2 \tau^2}{\sigma_d^2 \tau (1-\tau)}, \beta\right) \quad (5)$$

With $\sigma_d = 0.4$ at $\tau = 0.5$: $w_{PC}(0.5) = 7.25$ vs. RTC's $w_{RTC}(0.5) = 2.0$, a $3.6\times$ increase. Table I and Fig. 1 show the comparison.

TABLE I
GUIDANCE WEIGHT COMPARISON AT KEY TIMESTEPS ($\sigma_d = 0.4$, $\beta = 10$)

τ	w_{RTC}	w_{PC} (clipped)	w_{PC}/w_{RTC}
0.1	9.11	10.00	$1.10\times$
0.3	2.76	10.00	$3.62\times$
0.5	2.00	7.25	$3.63\times$
0.7	2.76	5.01	$1.82\times$
0.9	9.11	9.69	$1.06\times$

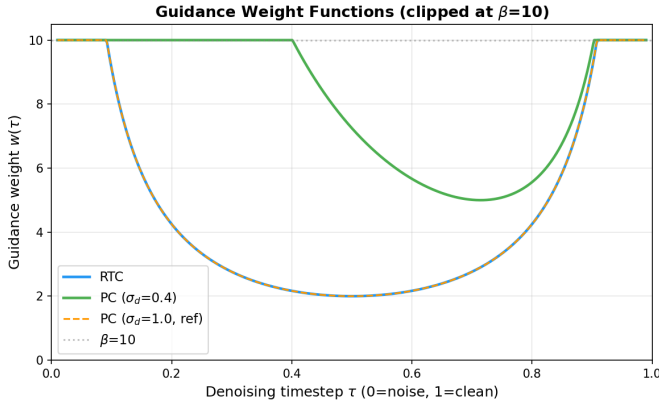


Fig. 1. Guidance weight function comparison. The prior-corrected weight ($\sigma_d=0.4$) is substantially higher than RTC at intermediate timesteps ($\tau=0.4-0.7$), strengthening correction during the middle denoising phase. At $\sigma_d=1.0$, the formula reduces to a profile close to RTC.

B. Orthogonal Trust-Region Constraint

Boosting guidance weights amplifies both parallel and perpendicular components. The perpendicular component causes abrupt direction changes at chunk boundaries. We constrain it via:

$$\max_{\hat{\mathbf{g}}} \langle \hat{\mathbf{g}}, \mathbf{g}_{PC} \rangle \quad (6a)$$

$$\text{s.t.} \quad \|\hat{\mathbf{g}} - \mathbf{g}_{\parallel}\|_2 \leq \rho \|\mathbf{v}_\tau\|_2 \quad (6b)$$

where $\mathbf{g}_{\parallel} = \frac{\mathbf{g}_{PC} \cdot \mathbf{v}_\tau}{\|\mathbf{v}_\tau\|^2} \mathbf{v}_\tau$ and $\mathbf{g}_{\perp} = \mathbf{g}_{PC} - \mathbf{g}_{\parallel}$. The closed-form solution is:

$$\mathbf{g}_{\text{final}} = \mathbf{g}_{\parallel} + \min\left(\frac{\rho \|\mathbf{v}_\tau\|}{\|\mathbf{g}_{\perp}\|}, 1\right) \mathbf{g}_{\perp} \quad (7)$$

which bounds the perpendicular component by $\rho \|\mathbf{v}_\tau\|$. Fig. 2 illustrates the constraint.

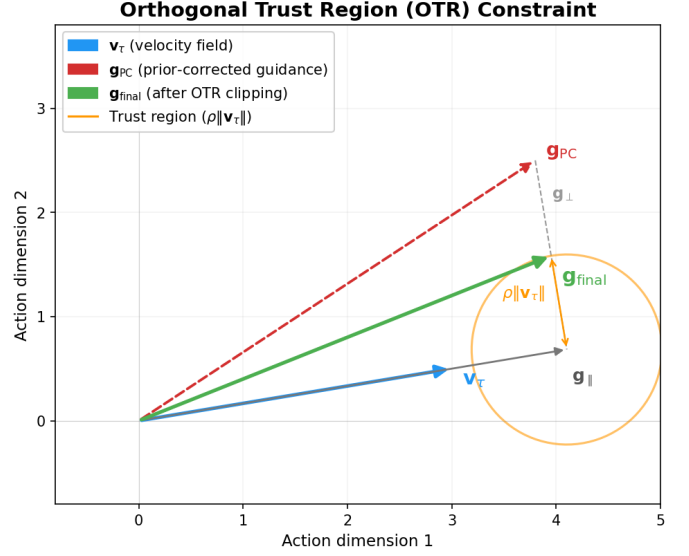


Fig. 2. Illustration of the OTR constraint. $\mathbf{g}_{PC}$ (red dashed) is the prior-corrected guidance vector, decomposed into a parallel component $\mathbf{g}_{\parallel}$ along $\mathbf{v}_\tau$ (blue) and a perpendicular component $\mathbf{g}_{\perp}$. The trust region (orange circle) is centered at $\mathbf{g}_{\parallel}$ with radius $\rho \|\mathbf{v}_\tau\|$. The final guidance $\mathbf{g}_{\text{final}}$ (green) retains the full parallel component while the perpendicular component is clipped to the trust-region boundary.

Remark 1 (Synergy between prior correction and OTR). *The two improvements are orthogonally complementary: the prior-corrected weight uniformly amplifies guidance strength along the correction direction, while the orthogonal trust-region constraint selectively clips the perpendicular component exceeding the trust region. Using prior-corrected weights alone amplifies the perpendicular component, exacerbating directional perturbation (risky); using the orthogonal trust region alone is limited in effectiveness when the original weight is already low; only their combination achieves both strong guidance and directional stability. The ablation study (Section IV-G) quantitatively validates this synergy across all 6 metrics.*

C. Complete Algorithm

Algorithm 1 presents the complete implementation of the proposed method within an Euler denoising solver.

IV. EXPERIMENTS

A. Experimental Setup

Benchmark. Experiments are conducted on the LIBERO [14] benchmark, which comprises 5 task suites: LIBERO-10 (10 tasks), LIBERO-Goal (10 tasks), LIBERO-Object (10 tasks), LIBERO-Spatial (10 tasks), and LIBERO-90 (90 tasks), totaling 130 distinct manipulation tasks.

Policy model. We use the open-source LIBERO checkpoint from $\pi_{0.5}$ [15] as the base policy model, employing the flow

Algorithm 1 POTR Guided Denoising: Prior-corrected Orthogonal Trust-Region (Euler Solver)

Require: Trained velocity field model v_θ ; initial noise $\mathbf{A}^0 \sim \mathcal{N}(0, \mathbf{I})$; observation $\mathbf{o}$; inpainting target $\mathbf{Y}$ (residual actions from previous chunk); soft mask $\mathbf{W}$; parameters σ_d, ρ, β , denoising steps n

Ensure: Clean action chunk $\mathbf{A}^1$

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1: for  $k = 0, 1, \dots, n-1$  do
2:    $\tau \leftarrow k/n$ 
3:    $\mathbf{v}_\tau \leftarrow v_\theta(\mathbf{A}^\tau, \tau, \mathbf{o})$   $\triangleright$  velocity field
4:    $\hat{\mathbf{A}}^1 \leftarrow \mathbf{A}^\tau + (1-\tau)\mathbf{v}_\tau$   $\triangleright$  one-step clean action estimate
5:    $\mathbf{g} \leftarrow (\mathbf{Y} - \hat{\mathbf{A}}^1)^\top \text{diag}(\mathbf{W}) \frac{\partial \hat{\mathbf{A}}^1}{\partial \mathbf{A}^\tau}$   $\triangleright$  pseudoinverse correction (VJP)
6:    $w \leftarrow \min\left(\frac{(1-\tau)^2 + \sigma_d^2 \tau^2}{\sigma_d^2 \tau(1-\tau)}, \beta\right)$   $\triangleright$  prior-corrected weight (Eq. (5))
7:    $\mathbf{g}_{\text{PC}} \leftarrow w \cdot \mathbf{g}$ 
8:    $\mathbf{g}_{\parallel} \leftarrow \frac{\langle \mathbf{g}_{\text{PC}}, \mathbf{v}_\tau \rangle}{\|\mathbf{v}_\tau\|^2} \mathbf{v}_\tau$   $\triangleright$  parallel component, fully retained
9:    $\mathbf{g}_{\perp} \leftarrow \mathbf{g}_{\text{PC}} - \mathbf{g}_{\parallel}$   $\triangleright$  perpendicular component
10:   $s \leftarrow \min\left(\frac{\rho \|\mathbf{v}_\tau\|}{\|\mathbf{g}_{\perp}\| + \varepsilon}, 1\right)$   $\triangleright$  OTR scaling ( $\varepsilon$ : numerical stability constant)
11:   $\mathbf{g}_{\text{final}} \leftarrow \mathbf{g}_{\parallel} + s \mathbf{g}_{\perp}$ 
12:   $\mathbf{A}^{\tau+1/n} \leftarrow \mathbf{A}^\tau + \frac{1}{n}(\mathbf{v}_\tau + \mathbf{g}_{\text{final}})$   $\triangleright$  Euler step
13: end for
14: return  $\mathbf{A}^1$ 
```

matching framework with prediction horizon $H = 10$ and an $n = 10$ -step Euler solver for denoising.

Evaluation protocol. Following the RTC [8] evaluation framework, we evaluate each method across 6 delay levels (delay $d = 0, 1, 2, 3, 4, 5$). The replan step is set to $s = \max(d, 1)$, i.e., delay 0 still replans every 1 step (fully closed-loop), while delay $d \geq 1$ replans every d steps. This yields $130 \times 6 = 780$ task configurations per method, and 3,120 total task configurations across all 4 methods.

Hyperparameters. Clipping threshold $\beta = n = 10$ (equal to the number of denoising steps). Rationale: the Euler solver scales the velocity field by $1/n$ at each step (Eq. (2)), so the effective correction strength is proportional to $1/n$; scaling β with n maintains consistent guidance across different denoising resolutions. Combined with the weight boost from $\sigma_d < 1$, $\beta = n$ clips extreme values while preserving sufficient correction headroom at intermediate timesteps. Data scale parameter $\sigma_d = 0.4$, trust-region radius ratio $\rho = 0.5$ (corresponding to $\|\mathbf{g}'_{\perp}\| \leq 0.5\|\mathbf{v}_\tau\|$). The detailed selection and grid search results for σ_d and ρ are presented in Section IV-C.

Compared methods.

- **Naive:** Baseline without RTC guidance.
- **RTC:** The original RTC method [8].
- **PC:** Prior-corrected weight only ($\sigma_d = 0.4$), without the orthogonal trust region, serving as an ablation baseline.
- **POTR (Ours):** The complete method (prior-corrected weight + orthogonal trust region).

B. Evaluation Metrics

We employ 6 metrics for comprehensive evaluation:

- **Task success rate** (success_rate $\uparrow$): Fraction of episodes in which the task is completed.

- **Average episode steps** (env_steps $\downarrow$): Mean environment steps per successful episode; lower indicates more efficient execution.
- **Chunk-switch L2 mean/max** (l2_mean/max $\downarrow$): L2 jump of the action vector at chunk boundaries, reflecting trajectory continuity.
- **Max acceleration/jerk** (max_acc/jerk $\downarrow$): Peak kinematic quantities in each episode’s action sequence, reflecting trajectory smoothness and safety.

Note: LIBERO operates in a normalized action space (not physical SI units). L2 jump, acceleration, and jerk are computed by finite differences on the normalized action sequence, and should be interpreted as relative smoothness indicators rather than physical quantities.

Aggregation. Cross-suite aggregation uses episode-weighted averaging:

$$\bar{m} = \frac{\sum_{s \in \mathcal{S}} N_s \cdot m_s}{\sum_{s \in \mathcal{S}} N_s} \quad (8)$$

where N_s is the number of tasks in suite s (LIBERO-90 has $N_{90} = 90$, the others $N_s = 10$ each), and m_s is the metric mean for suite s . Delay 0, which has no intra-chunk switching (L2 jump is identically 0), is excluded from the main tables; all reported metrics are arithmetic means over delay = 1–5.

C. Hyperparameter Selection

σ_d selection. We perform a grid search over $\sigma_d \in \{0.1, 0.2, 0.4, 0.6, 0.8, 1.0\}$ on all 5 task suites (delay $d=3$, replan $s=3$), using the 6 metrics as selection criteria. Table VIII reports the results.

$\sigma_d = 0.40$ achieves the best results on 3 of 6 metrics (l2_max, acc, jerk) and a near-optimal success rate (.464, within .023 of the best .487), striking the best balance between

TABLE VIII
 σ_d GRID SEARCH RESULTS (DELAY=3, PRIOR-CORRECTED WEIGHT, NO OTR)

σ_d	success $\uparrow$	steps $\downarrow$	l2_m $\downarrow$	l2_M $\downarrow$	acc $\downarrow$	jerk $\downarrow$
0.10	.487	293.6	.051	1.092	2.193	4.234
0.20	.449	301.7	.054	1.055	2.152	4.139
0.40	.464	299.6	.055	1.044	2.134	4.133
0.60	.474	296.1	.061	1.048	2.143	4.140
0.80	.487	295.6	.063	1.215	2.223	4.240
1.00	.477	295.4	.072	1.354	2.237	4.262

trajectory smoothness and task success rate. We therefore select $\sigma_d = 0.4$ as the default.

ρ selection. With $\sigma_d = 0.4$, $\beta = 10$, delay $d = 3$, and replan $s = 3$ fixed, we perform a grid search over the orthogonal trust-region radius ratio $\rho \in \{0.10, 0.25, 0.50, 0.75, 1.00\}$ on all 5 task suites. Table IX reports the results.

TABLE IX
 ρ GRID SEARCH RESULTS ($\sigma_d=0.4$, DELAY=3, POTR METHOD)

ρ	success $\uparrow$	steps $\downarrow$	l2_m $\downarrow$	l2_M $\downarrow$	acc $\downarrow$	jerk $\downarrow$
0.10	.462	299.9	.102	1.576	2.514	4.643
0.25	.487	294.4	.062	1.198	2.187	4.204
0.50	.531	288.2	.055	1.031	2.088	4.037
0.75	.495	294.0	.055	1.004	2.121	4.089
1.00	.480	296.0	.056	.965	2.091	4.047

$\rho = 0.50$ achieves the best results on 4 of 6 metrics (success, l2_mean, acc, jerk) and is near-optimal on l2_max (1.031 vs .965), striking the best balance between task success rate and trajectory smoothness. We therefore select $\rho = 0.5$ as the default, corresponding to $\|g'_\perp\| \leq 0.5\|v_\tau\|$.

D. Main Results

Table II presents the aggregated results of all 4 methods across all 5 task suites (averaged over delay 1–5).

POTR outperforms RTC on all 6 metrics:

- Task success rate improves from 0.495 to 0.520 (+5.1%), demonstrating that the improved guidance strategy not only enhances smoothness but also boosts task completion.
- Chunk-switch L2 max decreases from 1.446 to 1.120 (−22.5%), reducing trajectory discontinuity at chunk boundaries.
- Peak jerk decreases from 4.623 to 4.220 (−8.7%), indicating that the orthogonal trust region effectively suppresses action jitter.

Fig. 3 shows success rate across delay levels; POTR consistently outperforms RTC and Naive.

Worst-case analysis. Table III reports worst-case smoothness metrics: for each task suite, we first compute the mean over delay 1–5, then take the maximum across the 5 suites.

POTR achieves the best worst-case L2 jump (1.294) and jerk (4.348), with worst-case jerk 9.9% lower than RTC (4.826), demonstrating that POTR maintains advantage even on the most challenging task suites.

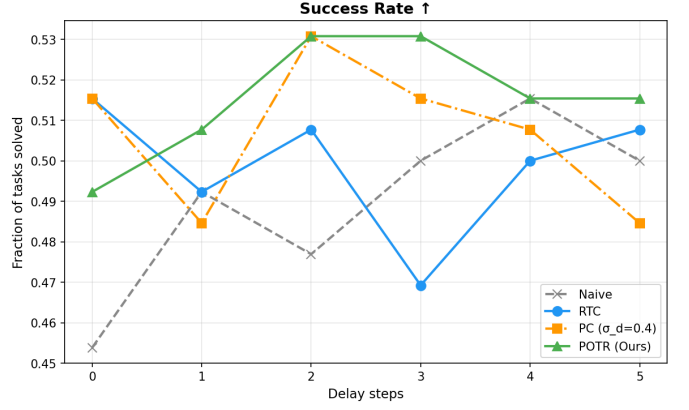


Fig. 3. Task success rate vs. delay level (5-suite episode-weighted). POTR (green) leads at most delay levels.

E. Per-Suite Analysis

Table V shows per-suite success rates. POTR achieves the largest improvement on LIBERO-90 (+10.7% relative), where task diversity makes chunk-boundary correction most impactful. On small suites (10 tasks), differences correspond to 1–2 episodes.

Table VI shows per-suite smoothness metrics.

F. Per-Delay Analysis

Table VII provides a detailed comparison between RTC and POTR at delay levels 1, 3, and 5.

POTR outperforms or matches RTC on 17 of 18 comparisons. Success rate improvement peaks at delay=3 (+13.2%); smoothness improvements are most pronounced at delay=1 (l2_M −19.4%, acc −14.4%, jerk −11.1%), where fast correction changes make directional constraints most effective.

G. Ablation Study

To verify the independent contribution of the orthogonal trust-region constraint, Table IV compares PC (prior-corrected weight only, $\sigma_d=0.4$) with the complete POTR method.

PC alone already outperforms RTC on all 6 metrics. OTR further improves success rate (+3.0%), acceleration (−3.4%), L2-max (−2.8%), and jerk (−2.4%), while L2-mean changes minimally (−0.2%). The directional constraint primarily suppresses extreme peaks.

H. Full Metric Overview

Fig. 4 shows the complete trends of all 6 metrics across delay levels.

V. CONCLUSION

We propose POTR for trajectory discontinuity in flow-matching robot policies. The prior-corrected weight (σ_d) strengthens intermediate-time guidance; the orthogonal trust-region (ρ) suppresses directional transients. On LIBERO (5 suites, 3,120 configurations), POTR outperforms RTC on all 6 metrics. Grid searches confirm $\sigma_d = 0.4$ (3/6 best) and $\rho = 0.5$ (4/6 best) with strong robustness across tested values.

TABLE II
MAIN RESULTS (EPISODE-WEIGHTED, DELAY 1–5 AVERAGE)

Method	success $\uparrow$	steps $\downarrow$	l2_m $\downarrow$	l2_M $\downarrow$	acc $\downarrow$	jerk $\downarrow$
Naive	.497	294.6	.224	1.921	2.989	5.483
RTC	.495	293.5	.083	1.446	2.518	4.623
POTR	.520	289.3	.064	1.120	2.226	4.220
vs RTC	+5.1%	−1.4%	−22.7%	−22.5%	−11.6%	−8.7%

Full Benchmark (5 suites, episode-weighted cross-suite aggregate)

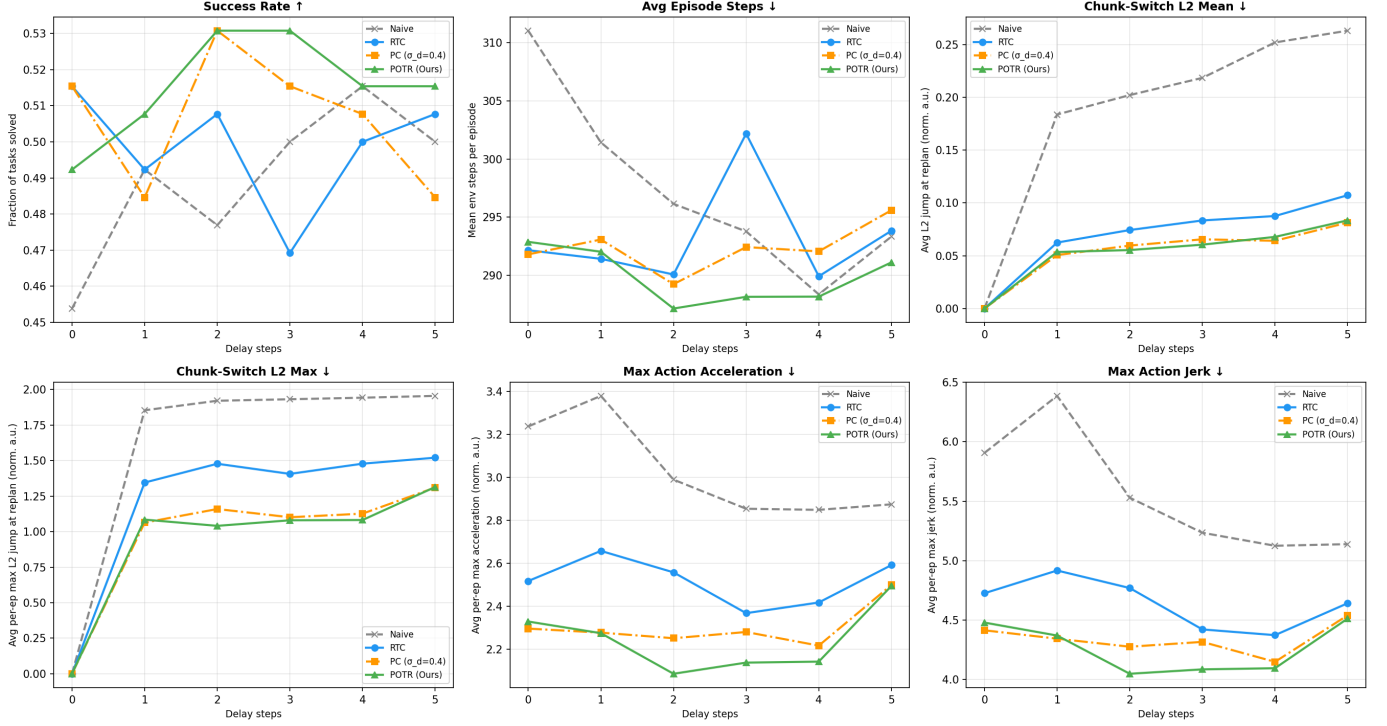


Fig. 4. All 6 metrics vs. delay level (5-suite episode-weighted aggregate). POTR (green solid line) achieves the best performance across all metrics and delay levels.

TABLE III
WORST-CASE SMOOTHNESS (SUITE MAXIMUM, DELAY 1–5 AVERAGE)

Method	worst_l2_m	worst_l2_M	worst_acc	worst_jerk
Naive	.263	2.016	3.232	5.850
RTC	.093	1.628	2.678	4.826
POTR	.070	1.294	2.327	4.348

TABLE V
PER-SUITE TASK SUCCESS RATE (DELAY 1–5 AVERAGE)

Task Suite	Naive	RTC	POTR
LIBERO-10	.940	.900	.900
LIBERO-Goal	.900	.960	1.000
LIBERO-Object	.960	1.000	1.000
LIBERO-Spatial	.980	.980	.980
LIBERO-90	.298	.289	.320

Limitations and future work. The proposed approach is in principle applicable to all action-chunking flow policies, but has so far been validated only on the $\pi_{0.5}$ [15] open-source

LIBERO checkpoint. The introduced σ_d and ρ serve as fixed hyperparameters whose optimal values may vary with task distribution and policy architecture. Future work will validate the method on different action-chunking flow policies and explore adaptive parameter selection strategies.

ACKNOWLEDGMENT

This work was supported by the National Key R&D Program of China (2023YFB4704900), the Scientific Instruments Development Program of NSFC (61527810), the Fundamental Research Funds for the Central Universities, the Aeronautical Science Foundation of China (20220056060001).

The authors are grateful to Key Laboratory of Autonomous Systems and Networked Control, Ministry of Education, Guangdong Engineering Technology Research Center of Unmanned Aerial Vehicle Systems, for their generous support of this research.

TABLE VI
PER-SUITE SMOOTHNESS METRICS (DELAY 1–5 AVERAGE, **BOLD** = BEST)

(a) Max Jerk ↓

Task Suite	Naive	RTC	POTR
LIBERO-10	5.75	4.44	4.25
LIBERO-Goal	3.86	3.42	3.46
LIBERO-Object	4.72	4.73	4.01
LIBERO-Spatial	4.30	4.07	4.01
LIBERO-90	5.85	4.83	4.35

(b) L2-Max ↓

Task Suite	Naive	RTC	POTR
LIBERO-10	1.94	1.28	1.01
LIBERO-Goal	1.46	0.79	0.59
LIBERO-Object	1.97	1.39	0.81
LIBERO-Spatial	1.47	0.69	0.52
LIBERO-90	2.02	1.63	1.29

TABLE VII
PER-DELAY BREAKDOWN (5-SUITE EPISODE-WEIGHTED)

Delay	Method	success ↑	steps ↓	l2_m ↓	l2_M ↓	acc ↓	jerk ↓
1	RTC	.492	291.4	.062	1.346	2.657	4.916
	POTR	.508	292.0	.054	1.085	2.274	4.368
3	RTC	.469	302.2	.083	1.407	2.367	4.420
	POTR	.531	288.2	.055	1.031	2.088	4.037
5	RTC	.508	293.8	.107	1.521	2.591	4.640
	POTR	.515	291.1	.083	1.314	2.494	4.511

TABLE IV
ABLATION: CONTRIBUTION OF THE ORTHOGONAL TRUST REGION (EPISODE-WEIGHTED, DELAY 1–5 AVERAGE)

Method	success ↑	steps ↓	l2_m ↓	l2_M ↓	acc ↓	jerk ↓
PC	.505	292.5	.064	1.152	2.304	4.323
POTR	.520	289.3	.064	1.120	2.226	4.220
Δ	+3.0%	−3.2	−0.2%	−2.8%	−3.4%	−2.4%

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